

Chapter 6: Anatomy of Flowering Plants

Monocot Stem

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Class Notes · Topic 5/8

Hypodermis = sclerenchyma

Ground tissue undifferentiated (no distinct cortex/pith)

Vascular bundles scattered, conjoint, closed, endarch

Bundle sheath of sclerenchyma around each VB

Phloem parenchyma absent; pith absent

Epidermis

Outermost single layer, compactly arranged barrel-shaped parenchyma cells covered with a thick cuticle.

Cortex — Hypodermis & Ground Tissue

Cortex is arranged in two sub-zones. **Hypodermis**: a few layers of **sclerenchymatous** cells just below the epidermis, providing mechanical strength. **Cortical layers (ground tissue)**: a large, conspicuous region of rounded, thin-walled parenchyma cells with intercellular spaces — cortex and pith are not clearly demarcated, so this whole undifferentiated region is called ground tissue.

Vascular Bundles

Numerous and **scattered** throughout the ground tissue, with peripheral bundles generally smaller than the centrally located ones. Each bundle is **conjoint and closed** (no cambium), with **endarch** xylem, and is surrounded by a **sclerenchymatous bundle sheath**.

Special Features

The **phloem parenchyma is absent** in monocot stems, and **water-containing cavities** are present within the vascular bundles (where the protoxylem elements have broken down). **Pith is absent**, since ground tissue is not differentiated into cortex and pith.

Dicot Stem vs Monocot Stem – Quick Contrast

Feature	Dicot Stem	Monocot Stem
Hypodermis	Collenchyma	Sclerenchyma
Vascular bundles	Ring, limited number	Scattered, numerous
Cambium / growth	Open, secondary growth occurs	Closed, no secondary growth
Pith	Present, well developed	Absent
Phloem parenchyma	Present	Absent

Monocot Stem at a Glance

- Sclerenchymatous hypodermis
- Scattered conjoint & closed vascular bundles
- Sclerenchymatous bundle sheath around each VB
- No pith, no phloem parenchyma